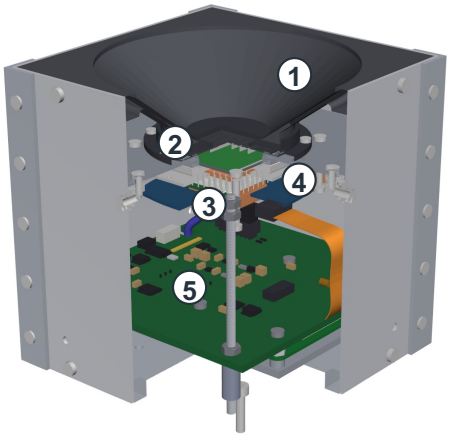
The payload and how it fits a host

Payload architecture, host requirements, and flight heritage

a SLANT payload



1 Baffle, matte black

2 Slit plates and five filters

3 Sensor and heat path

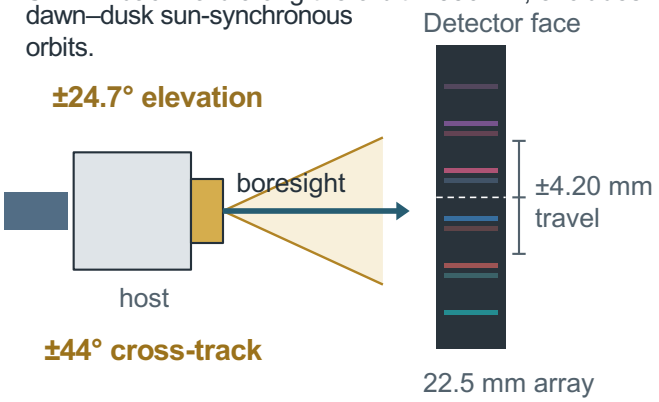
4 XIMEA sensor electronics

5 CM5 on carrier board

103 × 103 × 107 mm; modeled mass ≈665 g

b Acceptance cone

Three-axis-stabilized spacecraft; fixed bracket points SLANT backward along the orbit. ≥350 km; excludes dawn–dusk sun-synchronous orbits.



2.76 mm sweep; ±51.4° cross-track alone.
At combined tilts, mask any clipped rows.

c Signal and host interfaces

Optical head

Slit plates, filters; sensor on the XIMEA sensor board

↓ 60-pin link

XIMEA controller

↓ USB 3

Carthage carrier board with CM5

Row sums, storage, host link; power switching, watchdog

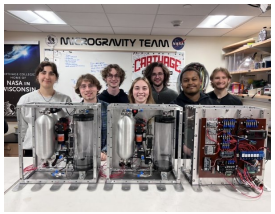
↓ Power: Glenair 801
↓ Data: 15-way micro-D

Host spacecraft

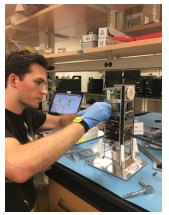
d Host requirements

Mounting	4 × M3 on a 66 mm square; conductive base; 800 g allocated. Mounting tolerance: ±24.7° elevation, ±44° cross-track. Keep ±40° × ±44° free of obstructions.
Thermal	Conductive heat path required. Baseplate −20 to +50 °C for survival, +20 °C or colder while acquiring. Detector uncooled.
Electrical	9–36 V, isolated; 5.7–9.7 W acquiring; 11.6 W allocated; off between passes; 0.33–0.55 W orbit average. Host switches power at times SLANT reports, or stays on.
Data and time	CAN 2.0B or RS-422; 7.2 MB/day (0.67 kbit/s mean); 1 PPS and UTC to 10 ms; position to 100 m; attitude knowledge 0.5°
Delivered	Interface control document: mounting, thermal, power, data, timing, keep-out; test reports; flight and ground software.
Not needed	Pointing service, radio, GNSS receiver, active cooling.

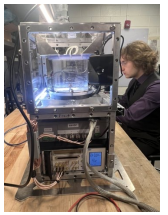
e Carthage payloads



NASA UPI Challenge: three identical payloads



CaNOP camera, engineering model



MUTT refueling, TechLeap award

22	2	6	5	2	1
Parabolic campaigns	ISS experiments	Suborbital flights	Sounding rockets	CubeSat missions	IM-3 lunar instrument

Host computer: Raspberry Pi CM5

Its predecessor, CM4, flies on SatGus and VERTECS.

Figure 4. (a) CAD layout and modeled mass. (b) Mounting tolerance: ±24.7° elevation and ±44° cross-track together, ±51.4° cross-track alone. (c–d) Host interfaces and requirements; allocations add 20 % to the CAD mass and estimated power. (e) Team flight heritage; flight heritage of compute board.